Full Stack Project Documentation with MERN

Project Documentation Format

1. Introduction

- **Project Title:** Automating Data Import and Relationship Mapping using Import Sets
- **Team Members:** S.Chitradevi

  A.Anishkumar

  V.Jeyasanthiya Arasu

  M.Deva

2. Project Overview

Purpose:

The purpose of this project is to automate the process of importing external data into the system and establishing relationships between records using Import Sets in ServiceNow. It reduces manual work and improves data accuracy.

Features:

- Automated data import from CSV/Excel
- Field mapping using Transform Maps
- Relationship creation between records
- Error handling and duplicate detection
- Data validation and integrity checks

3. Architecture

- **Frontend:**
  Uses the ServiceNow UI (forms, tables, and modules) to upload data and monitor imports.

- **Backend:**
  Business logic handled within ServiceNow using:
  - Import Sets
  - Transform Maps
  - Business Rules / Scripts
- **Database:**
  - Staging tables (Import Set tables)
  - Target tables (e.g., CMDB tables)
  - Relationship tables for linking records

4. Setup Instructions

- ### Prerequisites:
  - ServiceNow instance access
  - Admin or developer role
  - Sample CSV/Excel data
- ### Installation / Setup:

1. Log in to ServiceNow instance
2. Navigate to Import Sets → Load Data
3. Upload CSV/Excel file
4. Create Import Set table
5. Configure Transform Map
6. Run transformation

5. Folder Structure

Client Side:
Forms, UI pages, and modules

- **Server Side:**
    - Import Set Tables
    - Transform Maps
    - Script Includes / Business Rules

6. Running the Application

- Upload data file into Import Set
- Execute Transform Map
- Verify records in target tables
- Check relationships created

7. API Documentation

- Import Set API (for data loading)
- Table API (for accessing transformed data)

Example:

- GET /api/now/table/{table_name}
- POST /api/now/import/{import_set}

8. Authentication

- Authentication handled by ServiceNow
- Role-based access control (RBAC)
- Users require appropriate roles (admin/developer)

9. User Interface

- Data upload screen
- Import Set table view

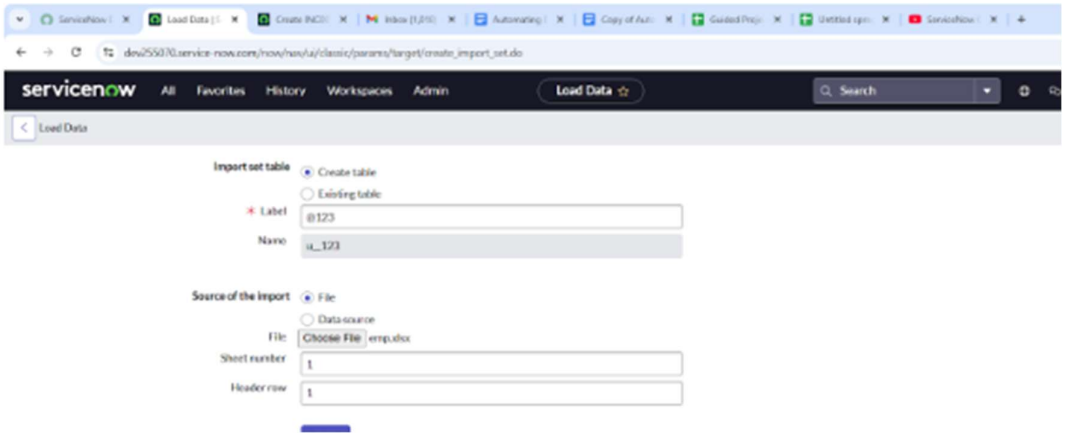
- Transform Map configuration page
- Target table records display

10. Testing

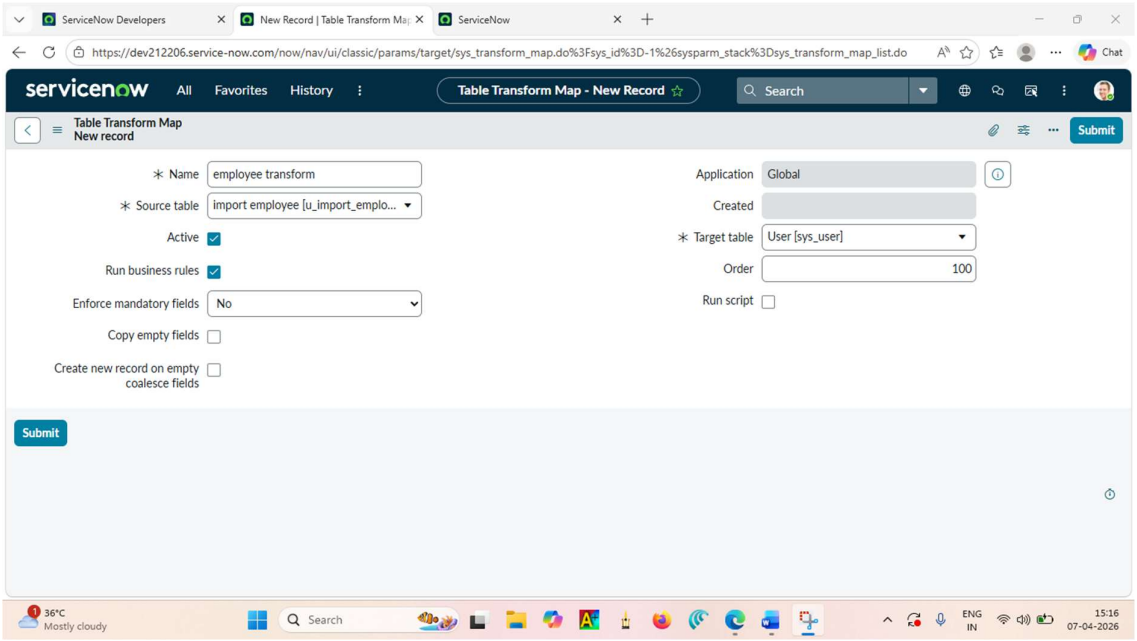
- Unit Testing for Transform Maps
- Data validation testing
- UAT for complete workflow
- Tools: ServiceNow test environment

11. Screenshots or Demo

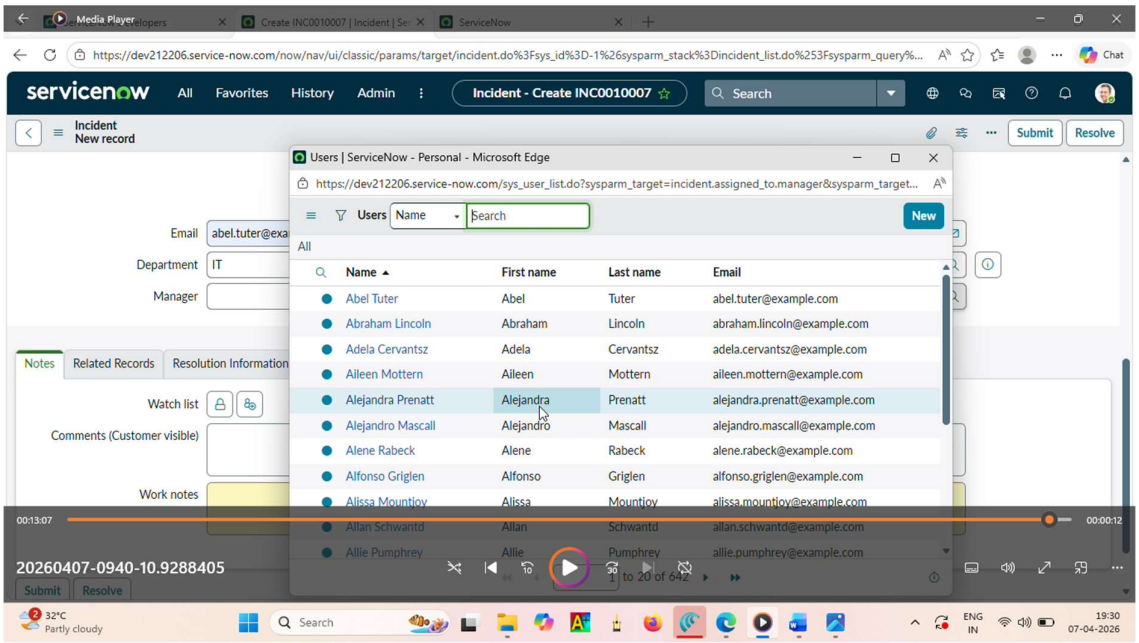
- Data upload



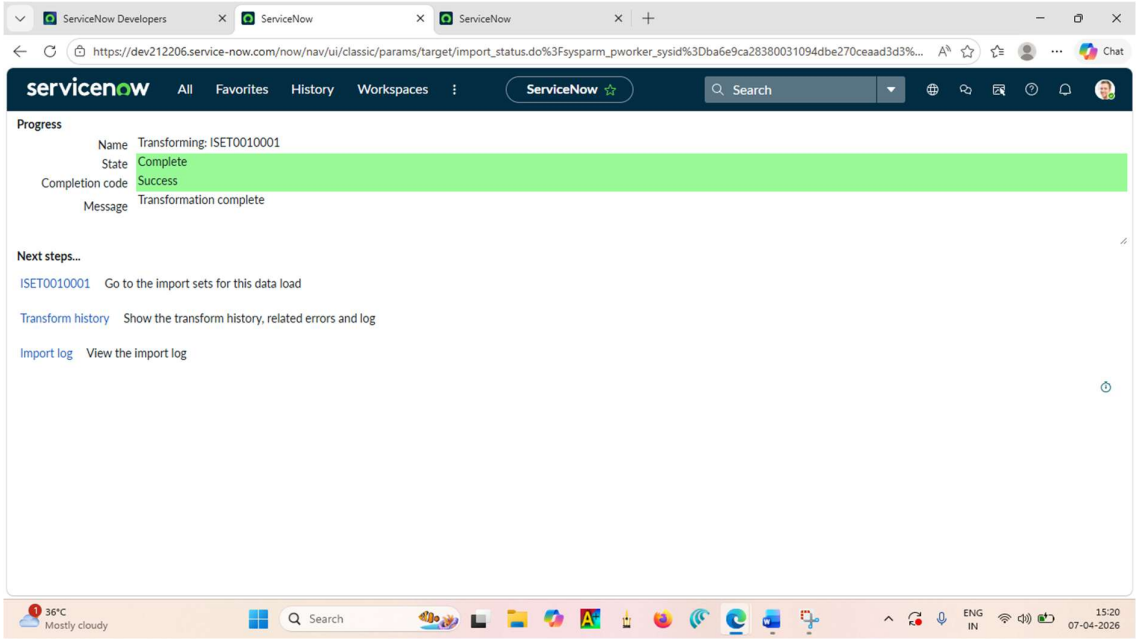
Transform Map



- Output tables



- Relationship mapping



12. Known Issues

- Data mismatch if mapping is incorrect
- Duplicate records if coalesce not set
- Errors during transformation for invalid data

13. Future Enhancements

- Real-time data integration using APIs
- Automated scheduling of imports
- Advanced error reporting dashboard
- AI-based data validation